

SUJAN SHARMA

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EDUCATION

TU Dortmund University

M.Sc. in Data Science (Pursuing)

Dortmund, Germany

Oct. 2025 – Present

Nepal Engineering College, Pokhara University

Bachelor of Computer Engineering | GPA 3.55/4.0 | Full Scholarship

Changunarayan, Bhaktapur

Nov. 2017 – Aug. 2022

WORK EXPERIENCE

Machine Learning Engineer

Fusemachines Nepal, Full-Time

Aug. 2023 – Nov. 2025

Kathmandu, Nepal

- Leveraged **Large Language Models** (GPT, Gemini, AWS Nova, Mistral) with **LangChain** and **LangGraph** for information retrieval; enforced strict **data definitions and consistency** in prompt engineering, improving chatbot response accuracy by **15%** and reducing inference costs by **20%** on election campaign based project.
- Developed **classification models** for action recognition and team classification in a **video analytics pipeline**, achieving **91% accuracy**; conducted **issue tracking and root cause analysis** on edge cases to reduce latency by 20%.
- Designed internal knowledge-sharing sessions on **Advanced RAG architectures**, emphasizing semantic chunking and **mapping logic** for retrieval; co-led few weekly research sessions bridging industry practice with applied research.
- Guided and evaluated 15 project teams at AI Fellowship Program 2025 as a Teaching Assistant; delivered 2 technical lectures as an Instructor.

AI Engineer

Letitu, Part-Time

Sept. 2023 – Dec. 2025

Remote – Korea

- Designed, trained, and deployed an end-to-end **ML classification pipeline** for student-college prediction, career mapping, implementing rigorous **data validation and quality checks** (80%+ accuracy), containerised with **Docker** and served on **AWS EC2**.
- Applied **SHAP** and **LIME** for model interpretability and feature importance analysis; automated the full model lifecycle via **CI/CD pipelines** ensuring reproducible training runs and versioned data artefacts.

Front-end Developer

Treeleaf Technologies, Full-Time

Nov. 2022 – Aug. 2023

Lalitpur, Nepal

- Contributed to building the data dashboard, share-web, and help-centre pages for the company's main product.
- Built a traffic violation management system for a public-sector client using Redux-Saga, protocol buffers, and WebSockets; followed Agile SDLC throughout.

TECHNICAL SKILLS

Data Governance & Quality: Data validation, Exploratory data analysis, data consistency checks, root-cause analysis, issue tracking, master data documentation

Databases & Data Eng.: PostgreSQL, MySQL, Snowflake, dbt, SQL, Python (Pandas, NumPy), ETL pipelines, FastAPI, Flask, RESTful APIs

ML / AI: PyTorch, TensorFlow, Scikit-Learn, MLflow, OpenCV, NLTK, Streamlit

LLM / GenAI: LangChain, LangGraph, Hugging Face, LLMs, RAG, Prompt Engineering, pgvector

Cloud & MLOps: Docker, AWS (EC2, SageMaker), CI/CD, MLflow, Git, GitHub Actions, Linux

Programming Languages: Python, JavaScript, C++, Bash

Communication: English (fluent), German (A1 - Learning)

PUBLICATIONS

Journal paper | Access

- Ojha, Trailokya Raj, . . . , **Sujan Sharma**. “Deep Learning CNN Models for Diseases Classification in Cauliflower Leaves.” *Journal of Artificial Intelligence and Capsule Networks* 7, no. 1 (2025): 32–50.

Conference paper | Ongoing

- Sujan Sharma**, Muskan Bhandari. “Cross-Modal Relational Knowledge Distillation for Efficient Rare Disease Patient Retrieval”

TEACHING EXPERIENCE

Visiting Faculty Lecturer

Jan. 2025 – Sept. 2025

St. Xavier's College (Tribhuvan University), Part-Time

Kathmandu, Nepal

- Taught **Information Retrieval** (7th sem.) and **Image Processing** (5th sem.) to 50+ undergraduates, blending theory with industry-relevant applications.
- Delivered a 24-hour hands-on training on **Generative AI and AI Agents** to 2nd- and 3rd-year students.

Visiting Faculty Lecturer

May 2023 – Oct. 2025

Vedas College (Tribhuvan University), Part-Time

Lalitpur, Nepal

- Taught 250+ undergraduates across **Discrete Structures, Computer Architecture, Computer Graphics, and Artificial Intelligence**; supervised 3 students on major projects.

TRAINING AND MENTORSHIP EXPERIENCE

Judge (online round) & Mentor

December 2025

Evaluated 18 projects and mentored 10+ teams during the 2-day IDEAX Hackathon at Madan B. Memorial College.

Generative AI Trainer

June 2025 – July 2025

Conducted a 45-hour training on Generative AI at Himalayan College of Engineering, covering all core concepts.

Python & ML Upskill Trainer

Dec. 2024 – March 2025

Delivered 3 months of advanced Python & ML training to 20+ undergraduates, focusing on core concepts and hands-on projects.

AI/ML Mentor

January 2024

Mentored 50 students in 5 groups on ML/AI, offering career guidance at an event organised by Prime IT Club.

Evaluator (Judge) in Tech Fest

December 2023

Judged 15+ Web and AI/ML projects aligned with SDGs at Sagarmatha Tech Fest, Sagarmatha Engineering College.

PROJECTS

Trustworthy Clinical Decision Support System | Research · Trustworthy AI for Healthcare

Feb 2026 – Mar 2026

- Built an agentic RAG system for medical QA over PubMed using Polars → PubMedBERT embeddings → Weaviate hybrid retrieval (BM25 + dense + filters), improving domain-specific retrieval quality.
- Implemented a LangGraph pipeline with re-ranking (MiniLM) and NLI-based verification, producing grounded answers with sentence-level explainability and interpretable trust scores for clinical use.

RAG-Based CV–Job Matching System | Python, LangChain, FastAPI, pgvector, Streamlit

Oct. 2025 – Nov. 2025

- Built an end-to-end Retrieval-Augmented Generation (RAG) pipeline for CV parsing, semantic similarity scoring, and explainable batch CV–job matching, served via a FastAPI backend with pgvector for efficient vector search.
- Implemented LLM-powered summarisation and prompt engineering strategies using LangChain and LangGraph; system supports batch processing with explainable match scores.

Major Project — Point Out Crops | Python, TensorFlow, Django, Flutter

Feb. 2022 – Aug. 2022

- Cauliflower disease classification on self-collected data with IoT-based real-time monitoring; achieved **93.47%** accuracy with ResNet50 after systematic model comparison.
- Hosted the classification model on Raspberry Pi to automate motor rotation and spray controls in real time.

CERTIFICATIONS & AWARDS

- Micro-degree in Artificial Intelligence – Fusemachines Jan 2024
- Generative AI with Large Language Models – DeepLearning.ai / Coursera Dec 2023
- Data Science Fellow – Fellowship.ai Jan 2023
- Full Scholarship, Bachelor of Computer Engineering – Nepal Engineering College (value > USD 8,000) 2017–2022